

6 (B)

$$u_1 = 15.0 \text{ ms}^{-1}$$



$$m_1 = 5.00 \text{ kg}$$

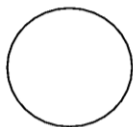
$$u_2 = 0 \text{ ms}^{-1}$$



$$m_2 = 10.0 \text{ kg}$$



v



$$m_f = 15.0 \text{ kg}$$

Using conservation of momentum,

$$m_1 u_1 + m_2 u_2 = m_f v$$

$$5.00(15.0) + 10.0(0) = 15.0(v)$$

$$v = 5.00 \text{ m s}^{-1}$$

$$\text{Loss of kinetic energy} = \frac{1}{2} m_1 u_1^2 - \frac{1}{2} m_f v^2$$

$$= \frac{1}{2} (5.00)(15.0)^2 - \frac{1}{2} (15.0)(5.00)^2$$

$$= 375 \text{ J}$$

7 (D)